Topology Optimization Front Suspension Upright of Electric Formula Car

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I. INTRODUCTION

The front upright stands as a pivotal component within racing suspension systems, serving as the anchor connecting essential elements like control arms, tie rods, and axles. Its role is vital as it smoothly transfers forces and moments between the wheel and control arm, ensuring a comfortable ride, steady steering, and easy wheel movement on different surfaces. In the competitive world of Formula Racing, where every fraction of a second matters, the front upright needs to be both strong and flexible. It must endure tough racing conditions while also being lightweight to save energy. Recent research has focused on making racing components lighter topology optimization of the front upright is crucial because it directly impacts the performance, safety, and efficiency of the racing car. By optimizing the design, it enhances strength, reduces weight, improves energy efficiency, and ensures structural integrity under intense racing conditions, ultimately contributing to better overall performance and competitive advantage in Formula Racing.

II. ENGINEERING DRAWINGS

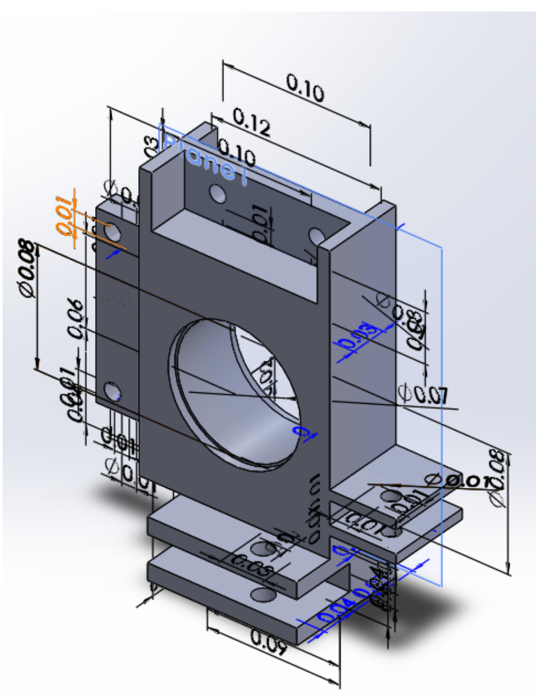


Fig. 1. Initial Design

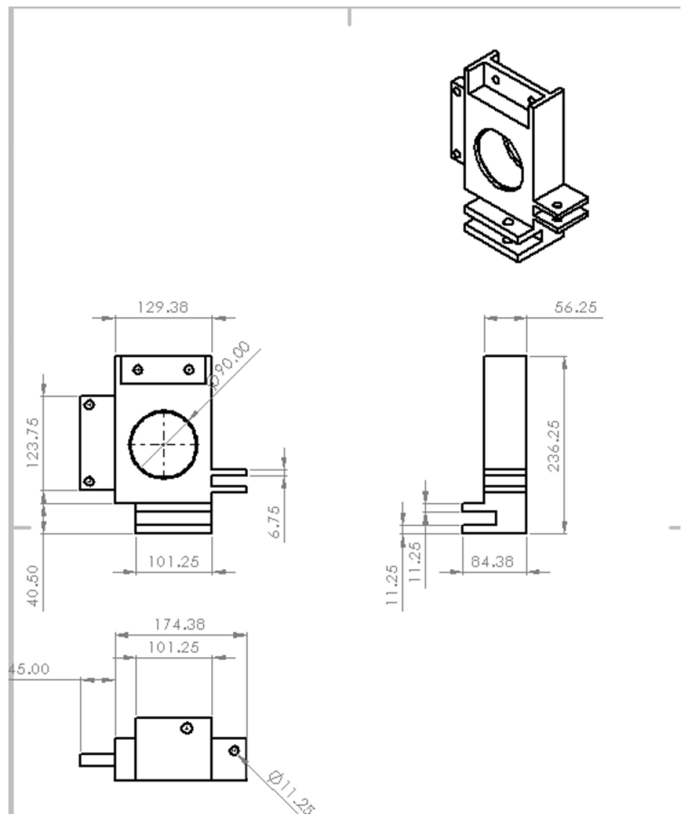
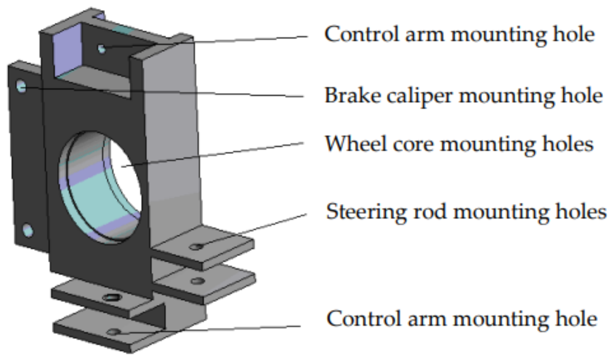


Fig. 2. 2D drawing of the initial design

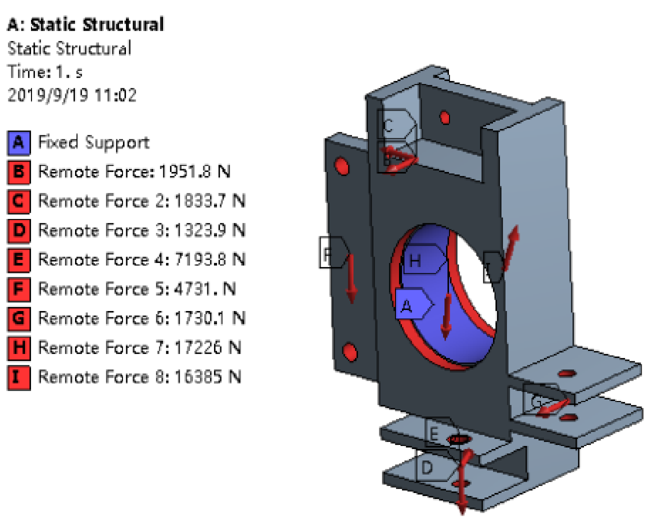
III. FORCE ANALYSIS

The preliminary design of the front upright is established through a meticulous process. It entails considering various factors such as the front wheel hub formula, the dimensions of inner and outer bearings, the inner wall size of the front upright, coordinates of mounting points for upper and lower control arms of the suspension, positioning of the steering tie rod, and the structural attributes of the front upright design, depicted in the following figure.



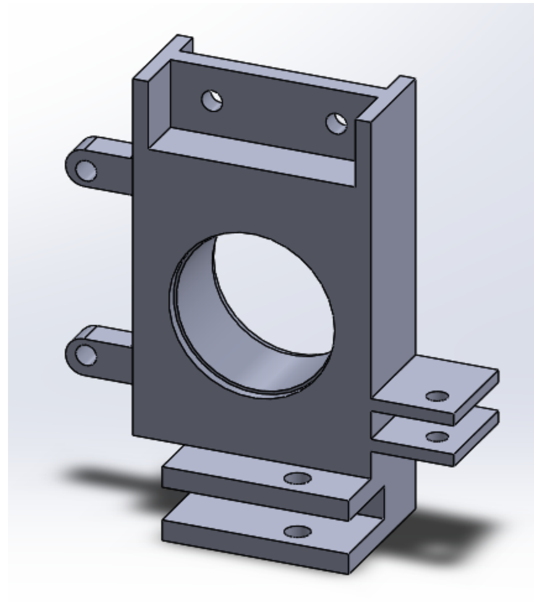
In adherence to competition regulations and the operational demands of the racing vehicle, it's understood that the front upright bears maximum loads during turning and braking maneuvers. Consequently, the initial front upright model undergoes topology optimization, specifically tailored to this demanding working scenario.

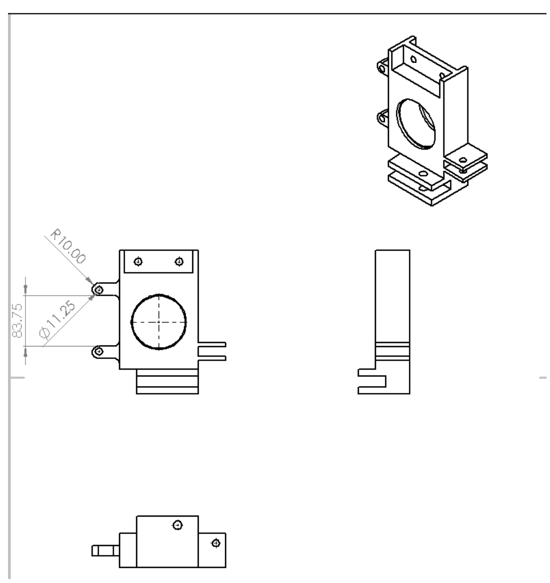
The dynamic loads on the moving caliper-mounting hole, as well as the inner and outer hub-bearing holes, are accurately applied based on predefined hard point coordinates. This encompasses meticulously determining both the magnitude and direction of each load acting on the system, as shown in the following.



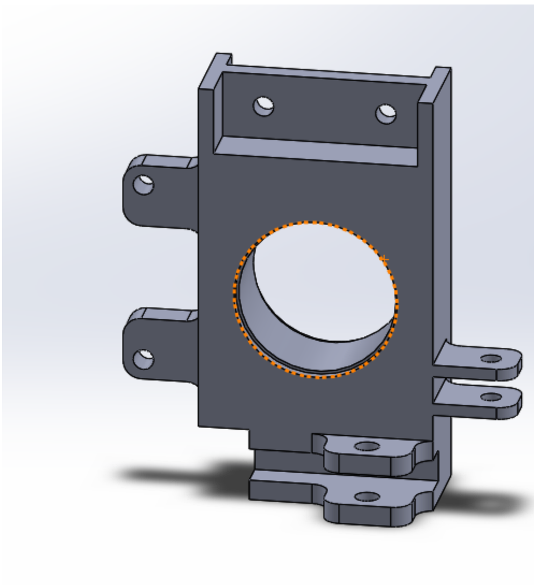
IV. Mechanical Modifications

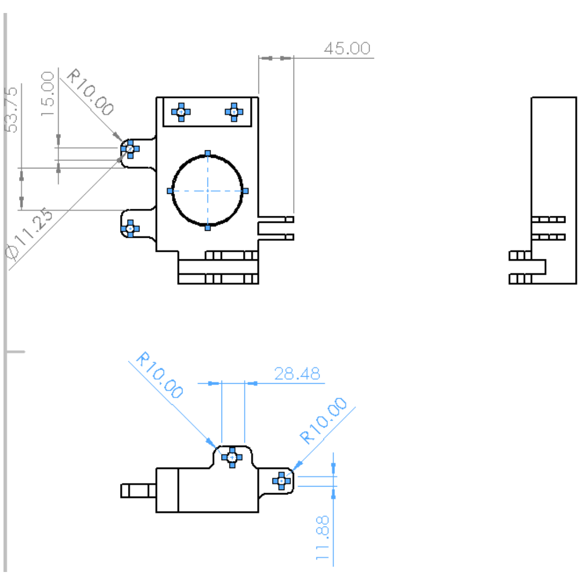
1) **First Modification**



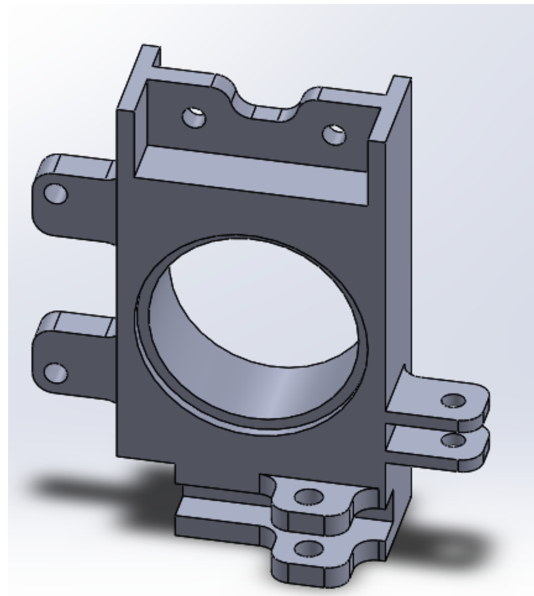


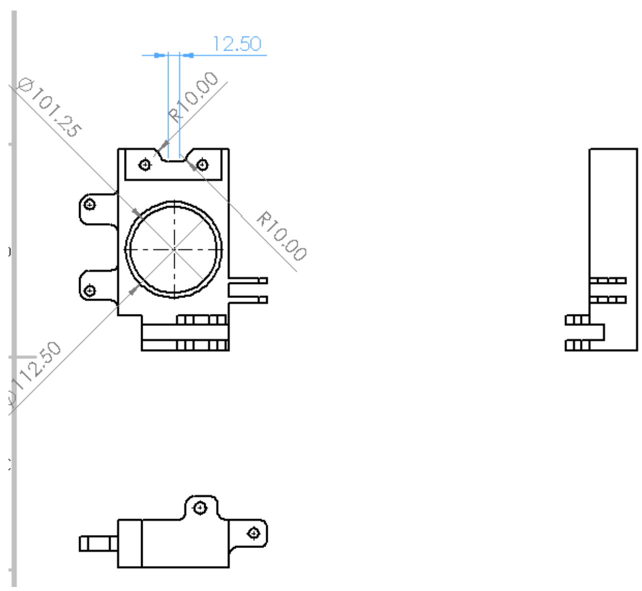
2) **Second Modification**



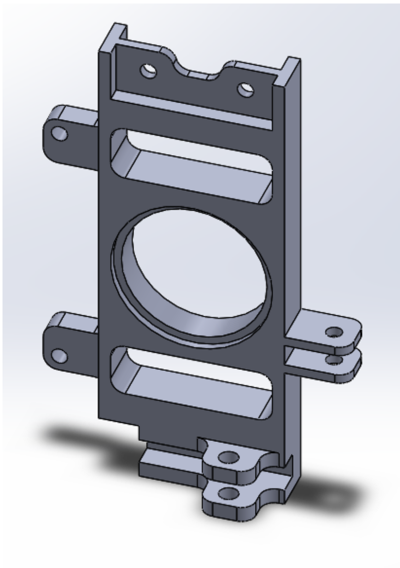


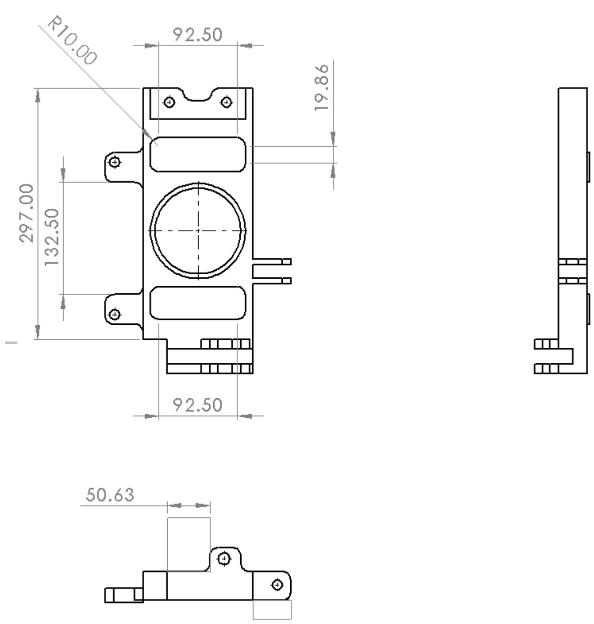
3) **Third Modification**



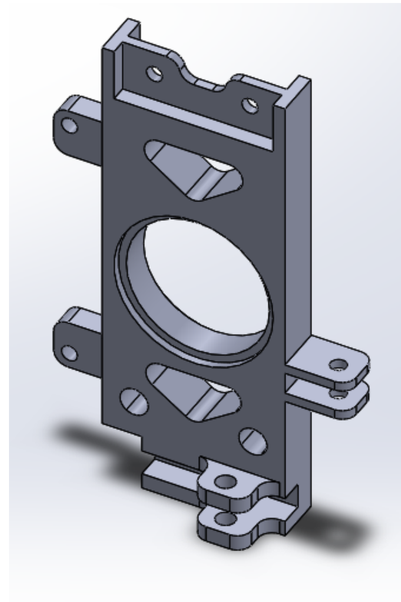


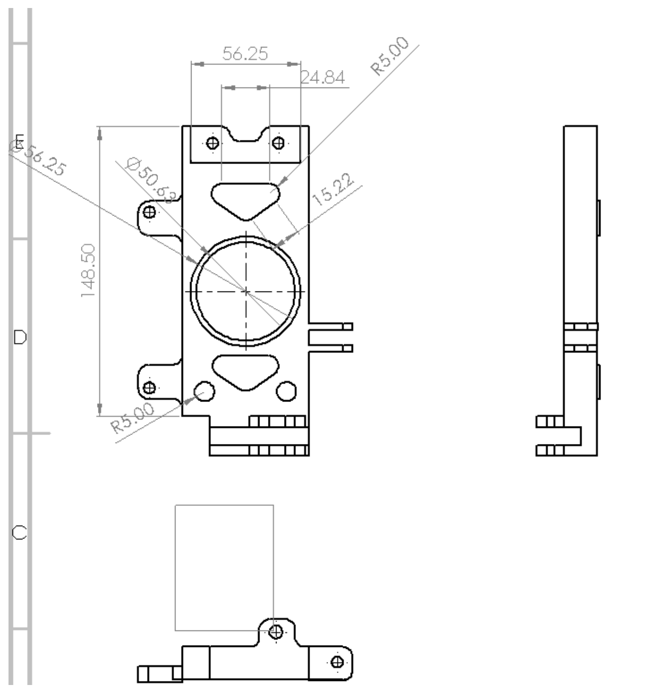
4) **Fourth Modification**





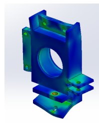
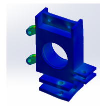
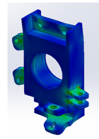
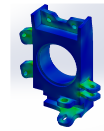
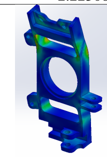
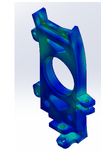
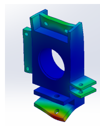
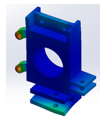
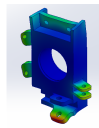
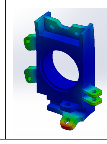
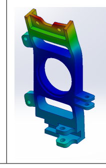
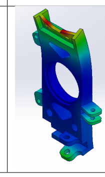
5) **Fifth Modification**





V. Simulation results analysis

The material used for simulation is 7075 aluminum, with Yield Strength 505 MPa

	Original	Mod1	Mod2	Mod3	Mod4	Mod5
Mass (*kg*)	3.60181	3.4231	3.35079	2.93006	2.22508	2.64046
Stress (*MPa*)	 0.03667	 124.8	 54.74	 57.12	 154.6	 84.29
Displacement (*mm*)	 5.781×10^{2}	 1.536×10^{1}	 7.034×10^{-2}	 7.8283×10^{-2}	 4.781×10^{-1}	 1.990×10^{-1}

VI. Decision Matrix

The chosen material is aluminum 7075-T6, priced at 202 LE/kg with 250 LE for manufacturing. Our analysis focuses on three key attributes: cost, weight, and safety factor. Our objective is to minimize the cost and weight while maximizing the safety factor.

Designs	Cost (L.E.) ↓	Weight (N) ↓	Safety Factor ↑
Initial Design	977.57	35.29	3.776
MOD 1	941.47	33.54	4.048
MOD 2	917.86	32.83	9.226
MOD 3	839.85	28.71	8.841
MOD 4	699.46	21.81	3.267
MOD 5	783.37	25.87	5.991

Choosing the initial design as the reference

Designs	Cost (L.E.) ↓	Weight (N) ↓	Safety Factor ↓	Total
Initial Design	1	1	1	3
MOD 1	0.96	0.95	1.072^{-1}	2.843
MOD 2	0.94	0.93	2.443^{-1}	2.279
MOD 3	0.86	0.81	2.341^{-1}	2.097
MOD 4	0.72	0.62	0.865^{-1}	2.496
MOD 5	0.80	0.73	1.587^{-1}	2.16

Considering the weight factors to be 0.4 for the cost, 0.4 for the weight and 0.2 for the safety factor

Designs	Cost (L.E.) ↓	Weight (N) ↓	Safety Factor ↓	Total
Initial Design	0.4	0.4	0.2	1
MOD 1	0.38	0.38	0.187	0.947
MOD 2	0.376	0.372	0.0189	0.7669
MOD 3	0.344	0.324	0.0854	0.7534
MOD 4	0.288	0.248	0.2312	0.7672
MOD 5	0.32	0.292	0.126	0.738

VII. Conclusion

Considering it is a loss decision matrix, the fifth modification stands out as the optimal choice.

VIII. references

References

[1] Jixiong Li, Jianliang Tan, and Jianbin Dong, *Lightweight Design of front suspension upright of electric formula car based on topology optimization method*, *World Electric Vehicle Journal*, vol. 11, no. 1, p. 15, 2020, MDPI.